

# Rova Isaia ANDRIAMAMY

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## Professional Objective

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Master's student in Artificial Intelligence and Astrophysics with experience in artificial intelligence, data analysis, and data science. Passionate about machine learning and deep learning, I enjoy designing practical data-driven solutions to solve real-world challenges.

## Projects and Experience

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### ML Model Deployment with Gradio

April 2026

Machine Learning, Data Analysis

Remote

[GitHub](#)

- Data analysis with NumPy, Pandas, Matplotlib and Seaborn
- Data preprocessing: cleaning, normalization, encoding and feature engineering
- Regression model building and evaluation (MSE, RMSE,  $R^2$ )
- Model deployment with Gradio on Hugging Face Spaces

### DARA Training - Radio Astronomy

July 2025 – March 2026

Python, Bash, CASA

South Africa / Kenya

[GitHub](#)

- Mastery of Bash scripts and Python libraries (NumPy, Astropy, SciPy, Matplotlib)
- Radio interferometry data reduction with CASA
- International collaboration with professionals

### Hack4dev International Hackathon

April 2025

Deep Learning, CNN

Anxian University, Antananarivo

[GitHub](#)

- Developed CNN architecture to classify CubeSat images into 5 categories
- Achieved 99.9% accuracy, 24x faster and 23x more compact than baseline
- Ranked in Top 20 internationally

### Food Price Prediction in Madagascar

2024 – 2025

Deep Learning, LSTM

M1 Project

[GitHub](#)

- Collected and cleaned 18+ years of market price data
- Created time-based features and tested RNN/LSTM/GRU models
- LSTM architecture achieving 94.51% prediction accuracy
- Developed interactive Flask/React dashboard for predictions

### BRICS Astronomy Project

May 2025 – July 2025

Machine Learning, Random Forest

Remote

[GitHub](#)

- Data analysis with Python (pandas, NumPy, Matplotlib, Seaborn)
- Random forest models to predict dark matter masses
- Achieved 96.8% accuracy with random forest regression
- Score of 89/100 in final project evaluation

## Education

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|----------------------------------------------------------------------------|-----------------------|
| <b>MASTER II</b><br>Artificial Intelligence, University of Antananarivo    | <i>2026 – Present</i> |
| <b>MASTER II</b><br>Astronomy and Astrophysics, University of Antananarivo | <i>2025 – Present</i> |
| <b>MASTER I</b><br>Artificial Intelligence, INSI University of Ambanidia   | <i>2024 – 2025</i>    |
| <b>BACHELOR’S DEGREE</b><br>Physics, University of Antananarivo            | <i>2022 – 2023</i>    |

## Technical Skills

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- **Programming Languages:** Python, C, R, MATLAB, HTML, CSS, JavaScript
- **Machine Learning:** scikit-learn (Linear Regression, Random Forest, XGBoost)
- **Deep Learning:** TensorFlow, PyTorch (CNN, LSTM, RNN, Object Detection)
- **Data Analysis and Visualization:** Pandas, NumPy, Matplotlib, Seaborn
- **Deployment:** Flask, React, Gradio
- **Tools and Technologies:** Git, GitHub, Docker, Linux, PostgreSQL, Power BI

## Key Certifications

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- [English for Career Development](#) - May 2026
- [ML Model Deployment with Gradio](#) - April 2026
- [Data Analysis with Power BI and AI](#) - March 2026
- [DARA Basic Training](#) (Data Reduction, Data Analysis) - March 2026
- [Introduction to Data Visualization](#) (Power BI, DataWrapper, Google Sheets) - February 2026
- [Fun Introduction to Coding](#) (HTML, CSS, Python, Data Science) - February 2026
- [Infrastructure for AI and Software Development](#) (Git, GitHub, Docker) - January 2026
- [British English Pronunciation Practice](#) - January 2026
- [DARA Computer Training](#) (Linux, Python) - August 2025
- [BRICS Astronomy Project](#) (ML, Data Analysis) - July 2025
- [American English Practice](#) - May 2025
- [Hack4dev Hackathon](#) (Deep Learning, CNN) - April 2025

## Affiliations / Associations

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- Malagasy Astronomical Society (MASS) – Participating Member
- Benty Fields Astronomy Research – Collaborative Research Group Member
- Association of Young Students of Sakay (AJES) – Educational Volunteer and Activity Facilitator
- ASCUT – University Sports and Cultural Association

## Additional Information

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- **Languages:** Malagasy (Native), French (Intermediate), English (Intermediate)
- **Interests:** Data Science, Data Analysis, Radio Astronomy